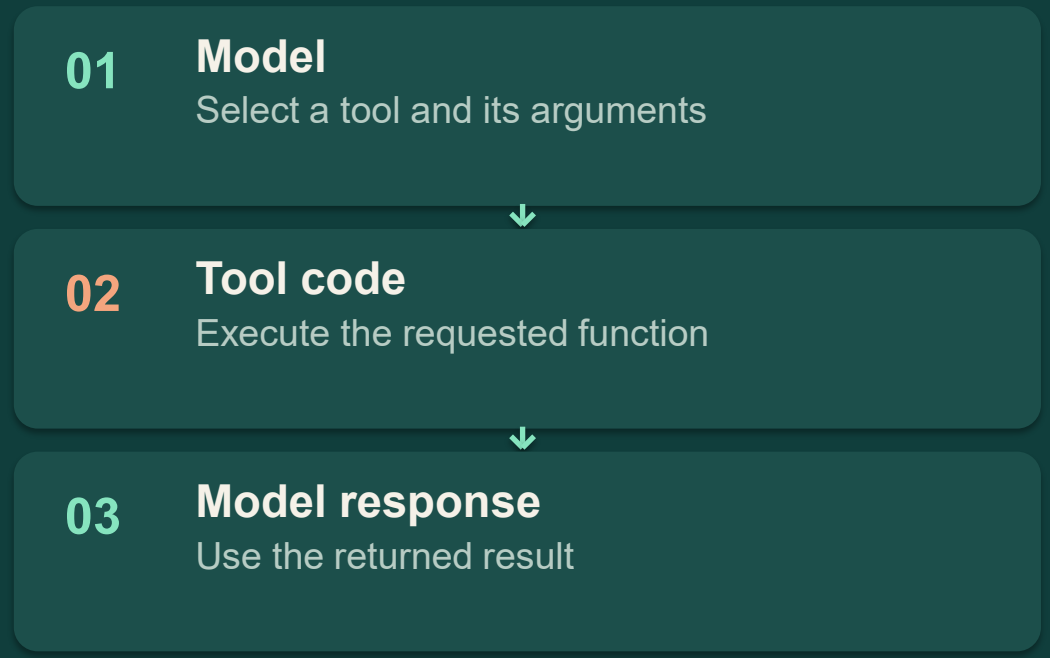
TOOL USE / FROM LANGUAGE TO ACTION

Give your agent tools

- Tools are executable code: functions or external API calls.
- The model selects a tool and supplies arguments.
- Execution produces a result the model can use.



01 / WHERE TOOL USE FITS

Start with the task

- Retrieve current information from APIs or databases.
- Execute code, generate reports or run simulations.
- Automate workflows and support customer queries.
- Use editing and evaluation tools for content.

01	**Retrieve**	APIs and databases
02	**Execute**	Code, reports and simulations
03	**Coordinate**	Workflows and customer support
04	**Edit**	Content checking and evaluation

Microsoft AI Agents for Beginners, Tool Use, lines 31–39

02 / AGENT IMPLEMENTATION

Build six connected components

- Schemas + execution logic describe and invoke tools.
- Message handling + integration connect the steps.
- Validation + state management keep interactions consistent.

01 **Describe + invoke**
Schemas and function execution logic

02 **Connect**
Message handling and tool integration

03 **Validate + remember**
Error handling and state management

Microsoft AI Agents for Beginners, Tool Use, lines 41–55

03 / FUNCTION SCHEMA

Write the tool’s contract

- Name: get_current_time
- Purpose: get the current time in a location.
- Parameter: location, a required string.
- A schema describes the function; code implements it.

01 **name**
get_current_time

02 **description**
Get the current time in a given location

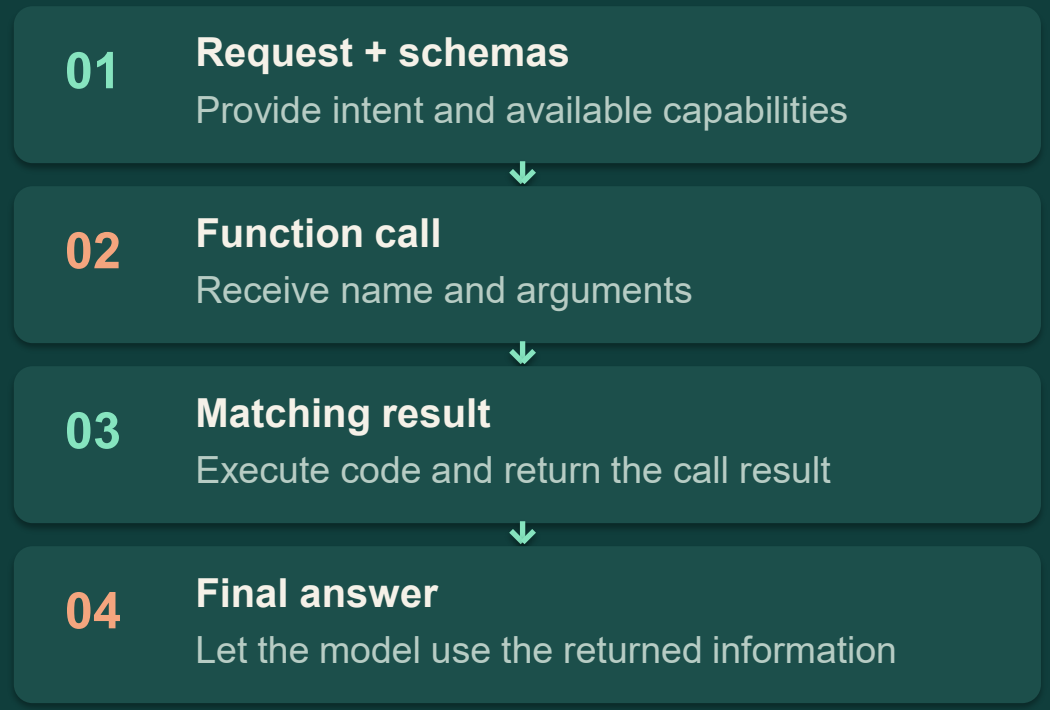
03 **parameters**
location: string · required

Microsoft AI Agents for Beginners, Tool Use, lines 63–108

04 / REQUEST → CALL → RESULT → ANSWER

Close the function-call loop

- Send the request and available function descriptions.
- Receive a function name and arguments.
- Execute the function; return its matching tool result.
- Let the model use that result to answer.



Microsoft AI Agents for Beginners, Tool Use, lines 61–192

05 / FRAMEWORKS AND MCP

Connect local and remote tools

- Decorated functions can become tools with generated schemas.
- A framework can handle the model–tool message exchange.
- An MCP server publishes tools and their schemas.
- Remote-tool queries are sent to that server.

01 **Local function**
Function → generated schema → model

02 **MCP server**
Published tools and schemas → framework

03 **Remote query**
The query is sent to the tool server

Microsoft AI Agents for Beginners, Tool Use, lines 208–245

06 / MANAGED AGENT SERVICE

Match tools to information or action

- Knowledge tools include search and file retrieval.
- Action tools include function calling and code interpretation.
- A toolset groups available tools.
- Threads retain the conversation’s message history.

01 **Knowledge tools**
Search and file retrieval

02 **Action tools**
Function calls and code interpretation

03 **Toolset / thread**
Available capabilities / message history

07 / TRUSTWORTHY TOOL USE

Bound database access

- Generated SQL can create injection or tampering risks.
- Configure restricted, read-only database access.
- Assign a read-only SELECT role where applicable.
- Use a secure execution environment.

01 **Generated SQL**
Recognise injection and tampering risks

↓

02 **Restricted role**
Read-only SELECT access

↓

03 **Protected execution**
Secure environment and read-only data

ORIGINAL QUESTION / FOLLOW THE CALL

What should the application do next?

- User asks for the current time in a city.
- Model returns get_current_time with location arguments.
- Is this already the final answer?
- Name the next two steps before revealing the solution.



Received

get_current_time(location = city)

Your decision

What must happen before the final answer?

ORIGINAL QUESTION / SOLUTION

Execute. Return. Then answer.

- Execute get_current_time using the supplied location.
- Send its result back with the matching call identifier.
- Let the model use the result to produce the final answer.
- The tool call and the tool result are different messages.

01 **Execute**
Run the selected function with its arguments

↓

02 **Return**
Attach the result to the matching call

↓

03 **Answer**
Use the returned information

Microsoft AI Agents for Beginners, Tool Use, lines 45–192